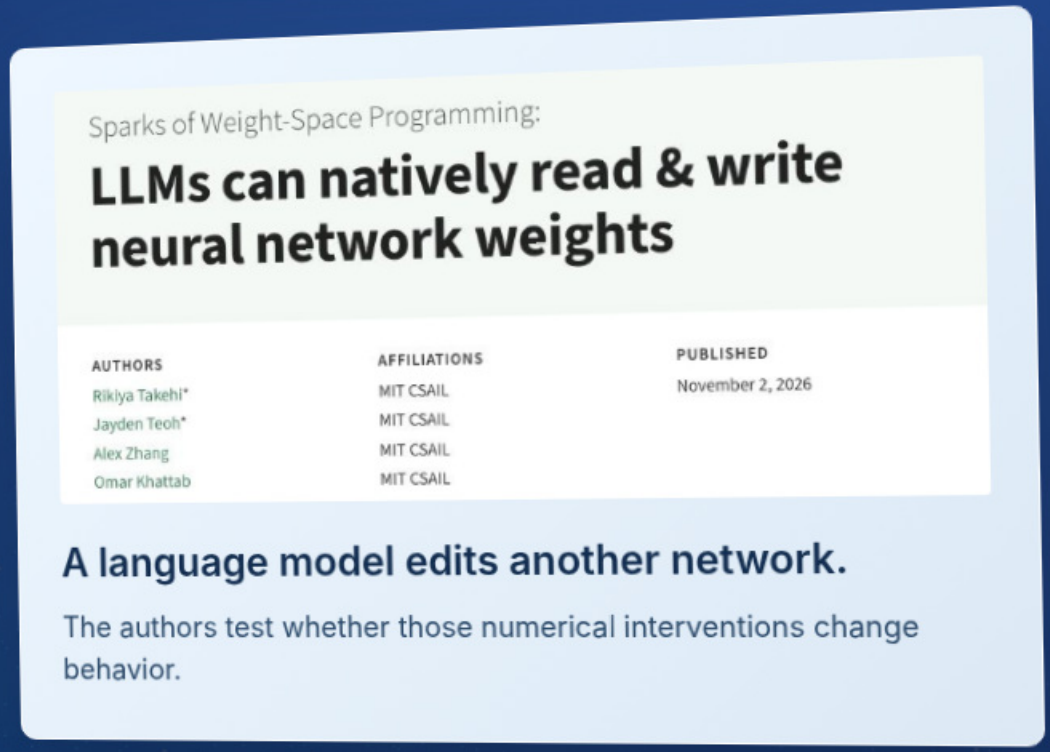
THE IDEA

Can one AI change how another AI behaves?

A research talk about reading model parameters, editing them, and measuring the consequences.



The authors tested the resulting behavior in games, control tasks and language models.

Source study · Takehi, Teoh, Zhang & Khattab · Sparks of Weight-Space Programming

THE SOURCE STUDY

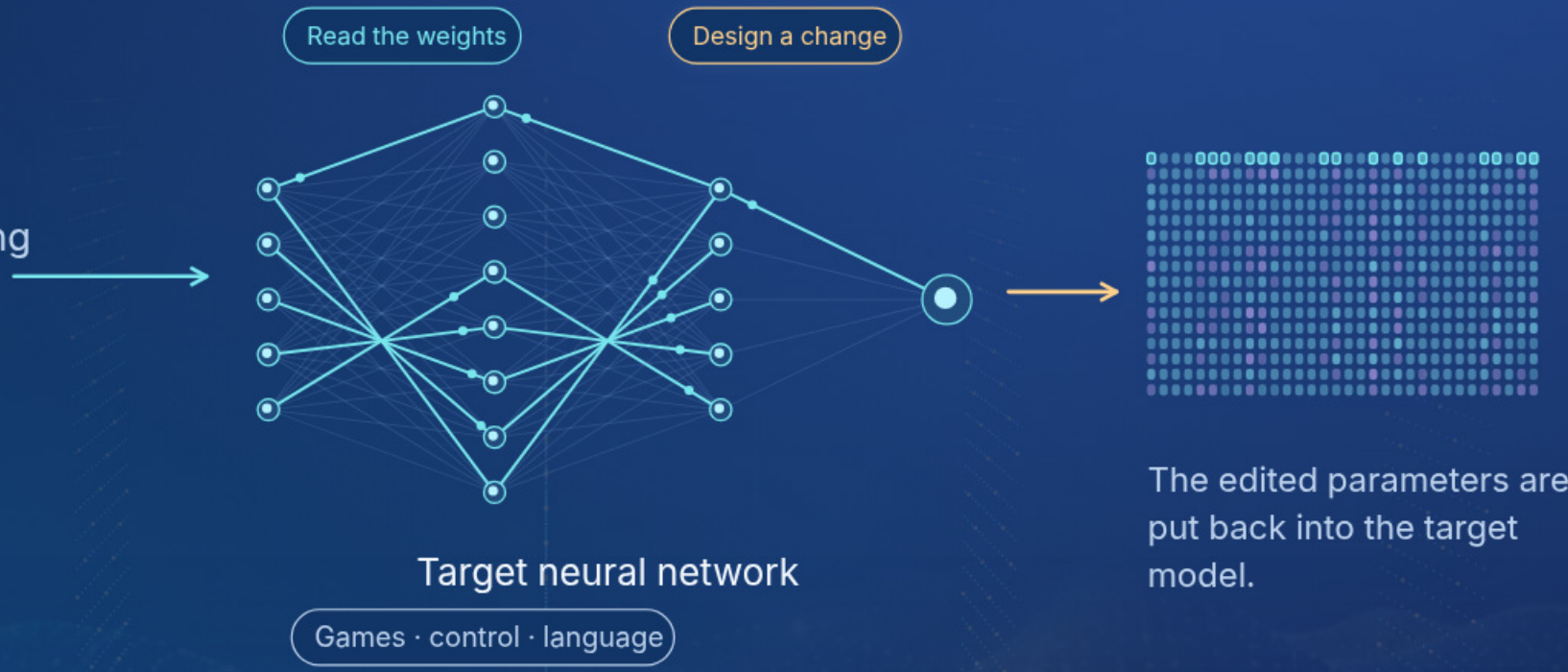
Astra used the task to design a weight change.

The source paper connects a behavior, the numbers that produce it, and an intervention.

GPT-6 Astra

Editor in the source study

Goal: change a target network's behavior by editing its parameters.



The edited parameters are put back into the target model.

Two examples show the idea clearly: a driving policy and a math-solving language model.

Source study · Takehi, Teoh, Zhang & Khattab · Sparks of Weight-Space Programming

SOURCE EXAMPLE · CONTROL

A stalled car gets a run-up.

Astra rewrote a small control path using the car's position and velocity.

The policy stalled

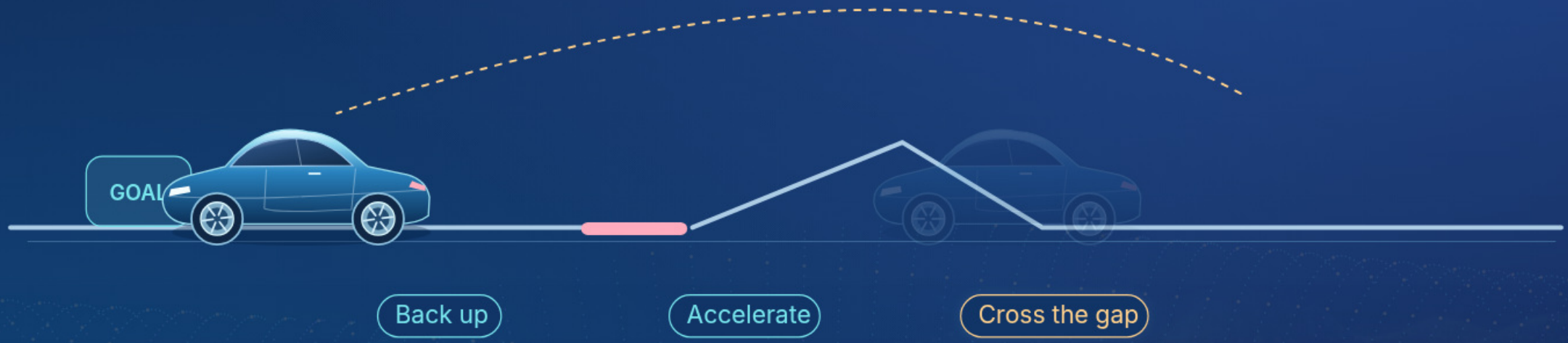
It repeatedly lost momentum on the ramp.

Astra's control path

Position + velocity → a run-up.

Measured result

1,024 successful test episodes out of 1,024.



> The edited policy reached the target in all 1,024 held-out test episodes.

Source study · Takehi, Teoh, Zhang & Khattab · Sparks of Weight-Space Programming

SOURCE EXAMPLE · LANGUAGE

A math model kept circling back.

Qwen3.5-0.8B often spent its output budget repeating its reasoning.

A math problem

The target must return a completed solution.

Qwen3.5-0.8B

The model generates reasoning before its final answer.



Reasoning returns to earlier content

85 / 96

Diagnostic answers reached the output limit.

Astra investigates the internal patterns associated with repetition.

Astra compares productive and repetitive contexts to locate the relevant contributions.

Source study · Takehi, Teoh, Zhang & Khattab · Sparks of Weight-Space Programming

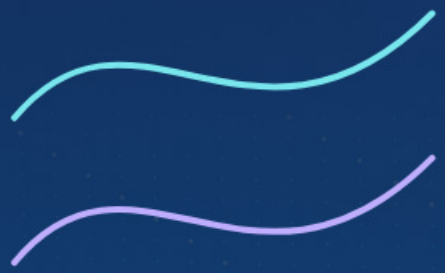
SOURCE EXAMPLE · THE EDIT

Reduce the contribution that drives repetition.

Astra measured a repetition-related direction and modified the MLP columns writing it.

1. Compare internal responses

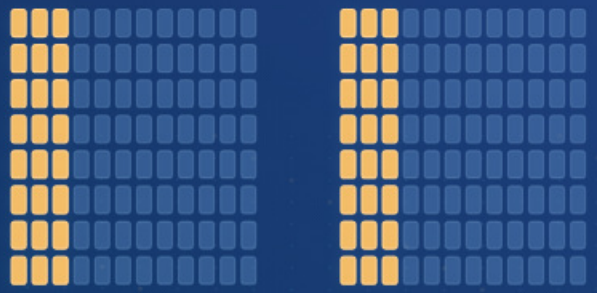
Productive context



Repetitive context

2. Target the strongest writers

Layer 15 Layer 19



64 columns per layer

3. Set the edit strength

6

Chosen multiplier for the repetition-direction correction

$$w' = w - 6(u^T w)u$$

u identifies the repetition-related contribution. The multiplier sets how strongly it is changed.

Strength exploration → selected 6

131,072 values targeted across the two matrices

> Each selected column has six times its projection along that direction subtracted.

Source study · Takehi, Teoh, Zhang & Khattab · Sparks of Weight-Space Programming

SOURCE EXAMPLE · OUTCOME

More answers reached a useful conclusion.

The same source evaluation settings were used before and after the edit.

Correct MATH-500 answers

Original → Edited

7.9% → 48.1%



Answers reaching the token limit

Original → Edited

93.0% → 33.1%



The measured gain accompanies fewer truncated generations.

> The authors' diagnosis links the gain to reducing repetitive generation.

Source study · Takehi, Teoh, Zhang & Khattab · Sparks of Weight-Space Programming

OUR STUDY

Three questions shaped our experiment.

We moved from the source examples to a census classifier, a compact language model and collaborative editing.

RQ1

Change the classifier

Can direct edits improve class-balanced accuracy after we remove connections?

RQ2

Improve answer selection

Can bounded changes improve SmolLM2 on science and physical-reasoning questions?

RQ3

Combine the editors

How does a feedback-informed joint session compare with independent editing?

RQ3 · How does feedback-informed collaboration compare with independent editing?

Seed 101 exploratory comparison · published snapshot d78bd5e · measured values from saved experiment records

OUR STUDY · DESIGN

Two targets. Four selected models.

Each editor starts from the same target. The joint session also receives the independent development histories.



Census-income model



SmolLM2-1.7B

GPT-6.1 Sol
Independent search

GPT-6 Astra
Independent search



Joint session

Previous patches + scores

Compare the four selected endpoints on the same evaluation records.

Starting model · Sol selection · Astra selection · joint selection.

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RQ1 · CENSUS CLASSIFICATION

A table of people and an income label.

UCI Adult / Census Income predicts annual income above or at most $50,000.

Age	Education	Occupation	Hours / week	Income label
39	Bachelors	Technical support	40	> $50,000
50	High school	Office work	40	≤ $50,000
28	Masters	Professional	45	> $50,000

48,842
records

14
original feature fields

The income label turns the record into a two-class prediction task.

The label is income above $50,000 or income at most $50,000.

UCI Adult / Census Income · illustrative table rows; published dataset size

RQ1 · PREPARATION

We trained the network, then opened an editing stage.

Each part of the data served a specific step in the experiment.

48,842 census records



32,561 training-source records

16,281 final test records

22,792
Fit the network

3,256
Calibrate the lesion

6,513
Build editing panels

104 encoded inputs → 32 ReLU → 16 ReLU → 1 sigmoid output

The remaining training data supplied lesion calibration, diagnostics and candidate feedback.

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RQ1 · A LIVE CALCULATION

A weight multiplies a signal.

Follow a worked numerical example from two inputs to an income-class decision.



First weight 0.200 0.400 0.200

Change the weight below: every dependent value and the decision update together.

Worked numerical example · scoring and arithmetic follow the stated operation

RQ1 · THE STARTING MODEL

We set 16 connection columns to zero.

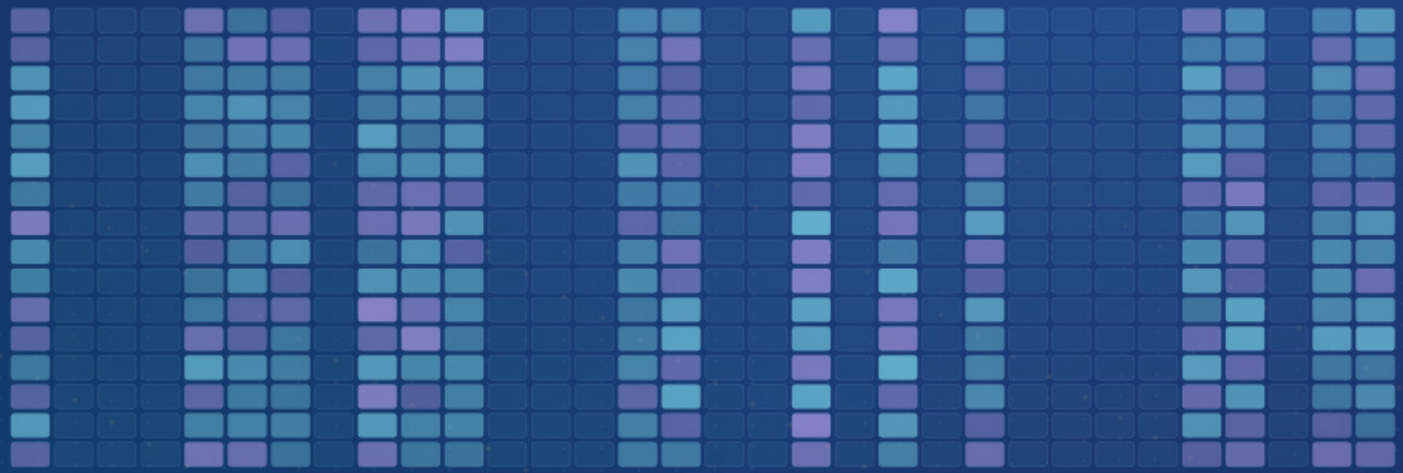
The editing task works on the remaining entries of the second-layer matrix.

Second-layer connections

32 incoming features × 16 destination neurons



Inspect a coefficient



16 active columns

16 columns held at zero

The editor sees the remaining learned coefficients and proposes changes to them.

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RQ1 · THE EDITOR'S BRIEF

Here is what we asked the GPT models to do.

A common instruction defines the goal, the information available and the editing actions.

Shared editor instruction
GPT-6.1 Sol · GPT-6 Astra

Improve the income classifier's development balanced accuracy.

Work on · The second-layer entries that remain active

Use · Selected-column scaling or explicit scalar additions

Select · Highest development score, then smallest change

Read the exact recorded prompt ↗

01 **Inspect**
Architecture, weights and 256 diagnostic records

02 **Propose**
One complete numeric patch to the allowed entries

03 **Evaluate**
A score on the 2,048-record development panel

The evaluator applies the proposed patch and returns measured feedback. The editor then chooses its next action.

Actions inspect the network, evaluate a complete patch and submit a selected candidate.

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RQ1 · THE TWO OPERATIONS

Add to a few entries. Or scale a column.

These two simple operations define the weight changes available to the editors.

Add to named entries

A sparse patch gives specific cells a correction.

0.20	-0.10	0.35	0.12
0.05	0.30	-0.17	0.25
0.14	-0.30	0.10	0.40

−0.20 + 0.03 = −0.17

Scale selected columns

The same factor multiplies a complete column.

0.200	-0.110	0.350	0.120
0.050	0.330	-0.200	0.250
0.140	-0.330	0.100	0.400

Column factor: ×1.10

The evaluator applies the candidate to the starting weights and returns its development score.

Seed 101 exploratory comparison · published snapshot d78bd5e · measured values from saved experiment records

RQ1 · GPT-6.1 SOL

Sol selected sixteen different additions.

The stored patch changes row 0 of the second-layer weight matrix.

GPT-6.1 Sol

Selected Adult intervention

16 entries → one destination neuron

column 0	column 4	column 5	column 6	column 8	column 9	column 10	column 14
-0.03094	-0.12239	0.04171	0.00464	-0.14811	-0.25384	0.35376	0.17289
+ 0.0315	+ 0.0518	+ 0.0280	+ 0.0252	+ 0.0358	+ 0.0346	+ 0.0532	+ 0.0417

column 15	column 18	column 20	column 22	column 27	column 28	column 30	column 31
0.15282	0.30316	-0.31110	0.20351	-0.12419	0.19956	0.15013	0.31604
+ 0.0280	+ 0.0116	+ 0.0290	+ 0.0280	+ 0.0307	+ 0.0211	+ 0.0488	+ 0.0472

Different magnitudes, applied together to the same row.

> The same patch is then applied across every record in the final evaluation.

Seed 101 exploratory comparison · published snapshot d78bd5e · measured values from saved experiment records

RQ1 · GPT-6 ASTRA

Astra selected +0.003 on each of sixteen entries.

Its recorded search compared several step sizes on the development panel.



+0.003

Selected equal addition

The +0.005 candidate tied on score. The smaller change won.

The selection rule chose the smaller +0.003 change.

Seed 101 exploratory comparison · published snapshot d78bd5e · measured values from saved experiment records

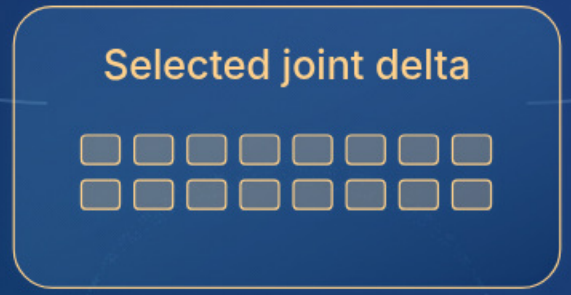
RQ3 · COLLABORATIVE EDITING

Proposal → review → measurement → next round.

The joint session uses both independent histories and alternates the two GPT roles.

GPT-6 Astra
Proposes

GPT-6.1 Sol
Reviews



$$\Delta W_{joint} = 0.93 \times \Delta W_{Sol}$$

Shared rounds 1 2 3 4 5

The selected Adult joint delta equals 0.93 × Sol's selected delta at all sixteen entries.

Seed 101 exploratory comparison · published snapshot d78bd5e · measured values from saved experiment records

RQ1 · THE SAME CONNECTION

One stored coefficient, three edits.

Compare the selected values for connection [0, 15], then multiply by the same activation.

Connection from hidden feature 15 to destination neuron 0



2.00 × 0.1508617 = 0.3017235

Activation 2.00

Starting | Astra | Sol | Joint

The joint patch adds 0.02604: 93% of Sol's addition.

Seed 101 exploratory comparison · published snapshot d78bd5e · measured values from saved experiment records

RQ1 · TWO RECORDED EXAMPLES

A small probability shift can cross the decision threshold.

These are saved predictions for two held-out records.



One decision becomes correct; the other becomes incorrect.

Seed 101 exploratory comparison · published snapshot d78bd5e · measured values from saved experiment records

RQ1 · ALL CHANGED DECISIONS

65 recovered decisions. 102 newly incorrect decisions.

For Sol's selected patch, the changes occur in different income classes.

Higher-income records

65 previously wrong decisions become correct.

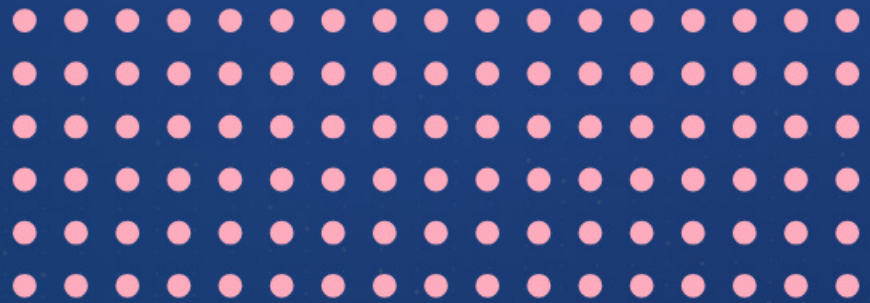


3,846

higher-income test records

Lower-income records

102 previously correct decisions become wrong.



12,435

lower-income test records

The test set contains 3,846 higher-income and 12,435 lower-income records.

Seed 101 exploratory comparison · published snapshot d78bd5e · measured values from saved experiment records

RQ1 · UNDERSTANDING THE RESULT

The two accuracy metrics weight records differently.

Ordinary accuracy counts records equally. Balanced accuracy averages the two class recalls.

Ordinary accuracy

Every record contributes equally.

65 – 102

16,281 records

−0.2273 pp

Balanced accuracy

Each income class contributes half.

65 / 3,846

– 102 / 12,435

Average the two class-rate changes

+0.4349 pp

Their equally weighted average gives +0.4349 percentage points of balanced accuracy.

Seed 101 exploratory comparison · published snapshot d78bd5e · measured values from saved experiment records

RQ1 · ANSWER

The class-balanced score increased.

Held-out evaluation of the four selected models · 16,281 records.

Starting	78.5858%
Sol	79.0207%
Astra	78.5855%
Joint	78.9668%



Ordinary accuracy

82.86 → 82.63%

Probability quality

Log-loss and Brier also increased for Sol and joint.

Ordinary accuracy and probability quality moved in the opposite direction for Sol and joint.

Seed 101 exploratory comparison · published snapshot d78bd5e · measured values from saved experiment records

RQ3 · ANSWER

The joint selection finished close to Sol.

The endpoint comparison links the cooperative outcome to the independent benchmark.

GPT-6.1 Sol

79.0207%

Balanced accuracy

Joint session

78.9668%

Balanced accuracy

Joint – Sol

−0.0539 pp



Ten-slot independent searches · five joint rounds with prior history

This seed compares ten-slot solo searches with five joint rounds using earlier feedback.

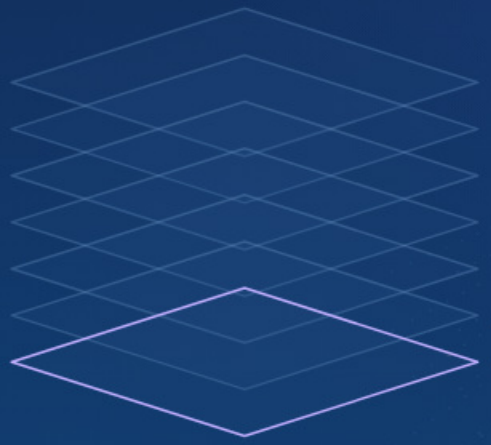
Seed 101 exploratory comparison · published snapshot d78bd5e · measured values from saved experiment records

RQ2 · A PRETRAINED LANGUAGE MODEL

The second target answers multiple-choice questions.

SmolLM2-1.7B starts from its pretrained weights.

SmolLM2-1.7B



Pretrained language model

ARC-Easy

School-science questions

Which material conducts electricity well? Copper · Wood · Glass · Rubber

PIQA

Physical commonsense questions

Choose the more effective way to accomplish a physical task.

The selected answer is determined by the benchmark's likelihood-scoring rule.

Seed 101 exploratory comparison · published snapshot d78bd5e · measured values from saved experiment records

RQ2 · THE EDITOR'S BRIEF

A clear objective and a bounded part of the model.

Both GPT editors receive the same language-model task and the same permitted operations.

Shared editor instruction

GPT-6.1 Sol · GPT-6 Astra

Improve SmolLM2's development answer-selection score.

Work on · MLP output weight matrices

Use · Selected-column scaling or explicit scalar additions

Select · Highest development score, then smallest change

Read the exact recorded prompt ↗

01 Inspect
Architecture, weights and 128 training diagnostics

02 Propose
One complete column-scale or scalar-addition patch

03 Evaluate
A score on the 128-example development panel

The evaluator applies the proposed patch and returns measured feedback. The editor then chooses its next action.

Inspect train diagnostics, propose a complete patch, read its score, then select a candidate.

Seed 101 exploratory comparison · published snapshot d78bd5e · measured values from saved experiment records

RQ2 · A SAVED CANDIDATE

One proposal halved 32 output columns.

A recorded candidate in transformer layer 23 changes the contribution of selected MLP features.

Transformer layer 23 · MLP output

Columns 0–31

Internal feature values



× 0.50

65,536

changed coefficients

75.00%

Development accuracy after normalization

Development acc_norm remains 75.00%, matching the starting score.

Seed 101 exploratory comparison · published snapshot d78bd5e · measured values from saved experiment records

RQ2 · HOW THE SCORE IS COMPUTED

Each answer receives a likelihood score.

A worked example shows the same scoring rule used by the benchmark.

Which material is a good electrical conductor?

Worked scoring example

Answer	Log-likelihood	Characters	Normalized score	
Copper	-1.20	6	-0.20	Chosen
Wood	-4.40	4	-1.10	
Glass	-5.00	5	-1.00	
Rubber	-4.80	6	-0.80	

The highest normalized score selects the answer; comparison with the key gives correctness.

Worked numerical example · scoring and arithmetic follow the stated operation

RQ2 · ANSWER

The search selected the original weights.

Five candidate attempts per independent editor; five joint attempts.



Selected change

0

All four endpoints use the original weights.

ARC-Easy 376 / 512 · 73.44%

PIQA 201 / 256 · 78.52%

The single selected weight state scores 73.44% on ARC-Easy and 78.52% on PIQA.

Seed 101 exploratory comparison · published snapshot d78bd5e · measured values from saved experiment records

THE FINDINGS

Three questions. Three measured answers.

The experiment connects proposed edits to actual weights, decisions and outcomes.

RQ1

Classifier

Sol gained +0.4349 pp in balanced accuracy.

The error distribution changed; ordinary accuracy fell.

RQ2

Language model

Every search selected the original SmolLM2 weights.

ARC-Easy 73.44% · PIQA 78.52%.

RQ3

Collaboration

The joint endpoint finished close to Sol.

Joint − Sol: −0.0539 pp, with an interval spanning zero.

RQ3 · The joint endpoint stayed close to the strongest independent endpoint.

Seed 101 exploratory comparison · published snapshot d78bd5e · measured values from saved experiment records

FUTURE DIRECTIONS

Make the next intervention more informative.

Use what this experiment revealed to choose the next experiments.

01 / NEXT EXPERIMENT

Mechanism-guided edits

Use activation evidence to choose directions and intervention sites.

From “where can we edit?” to “which behavior are we targeting?”

02 / NEXT EXPERIMENT

Decision-aware objectives

Optimize class balance alongside error costs and probability quality.

The outcome is a tradeoff that should be chosen deliberately.

03 / NEXT EXPERIMENT

Controlled collaboration

Match budgets and history access, then repeat independent sessions.

Measure which part of collaboration adds value.

Compare equal-budget histories and multiple independent sessions to isolate collaboration effects.

Seed 101 exploratory comparison · published snapshot d78bd5e · measured values from saved experiment records